1. Chemical Energetics

1.1. Exothermic & Endothermic Reactions

Chemical reactions involve energy transfer between the system (the chemical reaction) and its surroundings.

Exothermic Reaction	Endothermic Reaction
Heat energy is released into the surroundings	Heat energy is absorbed from the surroundings
Bond forming reactions	Bond breaking reactions
Surrounding temperature increases	Surrounding temperature decreases

Common examples of exothermic and endothermic reactions.

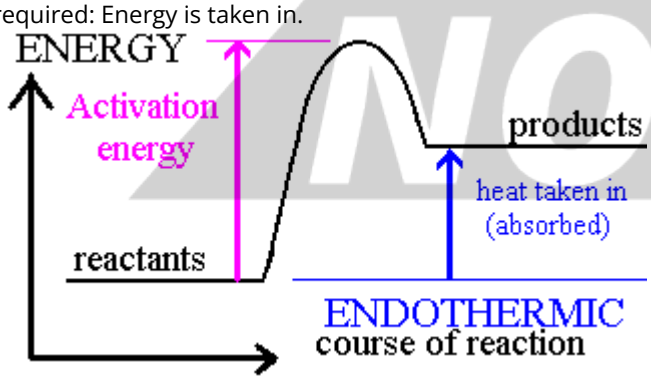
Exothermic Reaction	Endothermic Reaction
Combustion of fuels	Photosynthesis
Respiration	Thermal Decomposition
Neutralisation	Melting of Ice
Rusting of iron	Evaporation of water

1.2. Energy Level Diagrams

- Energy level diagrams represent chemical reactions that include the relative energies of the reactant and product.
- A reaction's energy change (ΔH) is represented by the difference in height between the reactant and its product.
- The activation energy (Ea) is the minimum energy required for the reaction to take place.

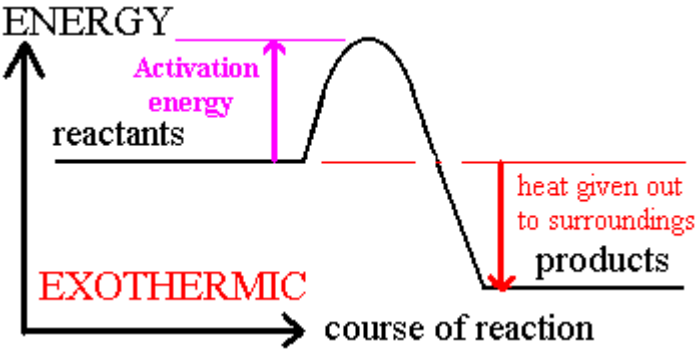
Endothermic Energy Level Diagram

- The system gains energy; higher activation energy is required: Energy is taken in.



Exothermic Energy Level Diagram

- The system loses energy; lower activation energy is required: Energy is given out.



1.3. Bond Energy

- This is the energy released when a bond is formed or required when a bond is broken respectively. The unit measure of this energy is kJ/mol.
- The energy change in a reaction is calculated using the following formula:

$$\Delta H = \text{Bond Breaking} - \text{Bond Forming}$$

- If the overall heat energy value is negative, the reaction is exothermic
- If the overall heat energy value is positive, the reaction is endothermic

Example

$H_2 + Cl_2 \rightarrow 2HCl$

Bond	Bond energy (kj/mol)
H - H	436
Cl - Cl	243
H - Cl	432

Bond breaking → 436 + 243 = 679 kj/mol

Bond forming → 2 (432) = 864 kj/mol

Thus, **Δ H** → 679 - 864 = -185 kj/mol

The reaction is **exothermic**.

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